Technical Qualification Form

Mechanical

Question 1:

What is the type of the aircraft that will participate in the competition?
(**Fixed-Wing:** Generates the lift/carrying force required for flight through its wings.
VTOL: Performs vertical takeoff/landing and carries out flight through its wings.)
If you selected **Other**, please provide an explanation.

Question 2:

Introduce the aircraft that will participate in the competition.
(Please specify whether the aircraft to be used in the competition will be a ready-made model or an original design, and summarize the design in one paragraph — design, type, manufacturing, material, etc. Maximum **500 characters**.)

Question 3:

What is the takeoff weight of the aircraft?
(In grams)
Note: There is a maximum limit of **12 kg**.

Question 4:

What is the length of the aircraft?
(In mm)

Question 5:

What is the width of the aircraft?
(In mm)

Question 6:

What is the height of the aircraft?
(In mm)

Question 7:

What is the takeoff type of the aircraft?
If you selected **Other**, please provide an explanation.

Question 8:

Explain the aircraft's takeoff.
(For example: takeoff run distance, climb height, catapult dimensions, takeoff mode — Maximum **300 characters**.)

Question 9:

What is the landing type of the aircraft?
If you selected **Other**, please provide an explanation.

Question 10:

Explain the aircraft's landing.
(For example: parachute size, glide slope, fuselage protection system — Maximum **300 characters**.)

Question 14:

Briefly explain the propulsion system to be used.
(For example: motor type, motor/ESC brand and model, ESC maximum current, propeller size, maximum thrust — Maximum **300 characters**.)

Question 20:

Provide information on how the kamikaze mission will be carried out.
(Information should be given about the algorithms and controllers to be designed for kamikaze mission stages such as transit, dive, and go-around — Maximum **500 characters**.)

Question 21:

Provide information on how the approach and tracking mission toward the rival aircraft will be carried out.
(Information should be given about the algorithms and controllers to be designed for use in the Combat UAV mission. In response to errors in controller inputs, the UAV's guidance algorithm, the image processing method to be used, and the model algorithms should be explained theoretically — Maximum **500 characters**.)

Electrical-Electronics Team

Question 11:

Explain your system architecture.
(Upload a visual representation of the system architecture you will use to the system — e.g., in-vehicle avionics connections. The uploaded architecture is expected to indicate systems such as communication and power.)

Question 12:

What is the battery type of the aircraft?
(Provide information about the capacity, cell count, battery type, etc. of the power source that will supply electronic systems in the aircraft such as the autopilot and ESC. Example: **10000 mAh 4S Li-Po battery** — Maximum **300 characters**.)

Question 13:

Explain the aircraft's power system.
(For example: reason for selection, capacity, charging, estimated flight time — Maximum **300 characters**.)

Question 14:

Briefly explain the propulsion system to be used.
(For example: motor type, motor/ESC brand and model, ESC

maximum current, propeller size, maximum thrust — Maximum **300 characters**.)

Question 15:

Define your autopilot system.
(Explain the hardware and software features you will use for the autopilot system in the aircraft — Maximum **500 characters**.)

Question 16:

Briefly explain the imaging system to be used.
(For example: camera brand and model, camera resolution, camera field of view, camera type (analog/digital) — Maximum **300 characters**.)
Note: A minimum resolution of **640x480** is recommended.

Question 17:

Define your RF devices.
(To prevent frequency conflicts with other teams during the competition, teams will be assigned a certain frequency range. In order for frequency assignment to be carried out properly and effectively, it is mandatory to use RF devices with adjustable frequencies. Define the device brand and model, adjustable frequency range, and communication range. Maximum **300 characters**.)

Software Team

Question 11:

Explain your system architecture.
(Upload a visual representation of the system architecture you will use to the system — e.g., in-vehicle avionics connections. The uploaded architecture is expected to indicate systems such as communication and power.)

Question 15:

Define your autopilot system.
(Explain the hardware and software features you will use for the autopilot system in the aircraft — Maximum **500 characters**.)

Question 18:

Define the ground control station.
(Which software/programs are you considering using for the Ground Control Station? Explain — Maximum **500 characters**.)

Question 19:

Define the image processing and transmission systems.
(Provide information on how the image will be processed and on the image processing algorithm planned to be used — Maximum **500 characters**.)

Question 20:

Provide information on how the kamikaze mission will be carried out.
(Information should be given about the algorithms and controllers to be designed for kamikaze mission stages such as transit, dive, and go-around — Maximum **500 characters**.)

Question 21:

Provide information on how the approach and tracking mission toward the rival aircraft will be carried out.
(Information should be given about the algorithms and controllers to be designed for use in the Combat UAV mission. In response to errors in controller inputs, the UAV's guidance algorithm, the image processing method to be used, and the model algorithms should be explained theoretically — Maximum **500 characters**.)

Question 22: Provide information on the method to be followed for escaping no-fly zones.
(Information should be given on how the prohibited flight zones will

be interpreted by the aircraft, as well as on the supervisory controller and general avoidance algorithms that will guide the UAV — Maximum **500 characters**.)

Joint Inter-Team Work Questions

Question No.	**Topic**	**Responsible Teams**
11	System Architecture	Electrical (Hardware) & Software (Communication Flow)
14	Propulsion System	Mechanical (Propeller/Thrust) & Electrical (Motor/ESC)
15	Autopilot	Software (Embedded Software) & Electrical (Hardware Integration)
20–21	Missions	Software (Algorithm) & Mechanical (Flight Dynamics / Dive Capability)